Neurobiology of Disease 177 (2023) 105997

Contents lists available at ScienceDirect

Neurobiology of Disease

journal homepage: www.elsevier.com/locate/ynbdi

Cross-sectional proteomic expression in Parkinson's disease-related proteins in drug-naïve patients vs healthy controls with longitudinal clinical follow-up

Ilham Yahya Abdi[a,b,1], Michael Bartl[c,1], Mohammed Dakna[c], Houari Abdesselem[b], Nour Majbour[b], Claudia Trenkwalder[d,e], Omar El-Agnaf[a,b,2,*], Brit Mollenhauer[c,d,*,2]

[a] College of Health and Life Sciences, Hamad Bin Khalifa University (HBKU), Education City, Qatar Foundation, Doha, Qatar
[b] Neurological Disorders Research Center, Qatar Biomedical Research Institute (QBRI), Hamad Bin Khalifa University (HBKU), Qatar Foundation, Doha, Qatar
[c] Department of Neurology, University Medical Center Goettingen, Robert-Koch, Goettingen, Germany
[d] Paracelsus-Elena-Klinik, Klinikstrasse, Kassel, Germany
[e] Department of Neurosurgery, University Medical Center Goettingen, Robert-Koch, Goettingen, Germany

ARTICLE INFO

Keywords:
Parkinson's disease
Plasma proteomics
Biomarkers
SOMAscan assay

ABSTRACT

There is an urgent need to find reliable and accessible blood-based biomarkers for early diagnosis of Parkinson's disease (PD) correlating with clinical symptoms and displaying predictive potential to improve future clinical trials. This led us to a conduct large-scale proteomics approach using an advanced high-throughput proteomics technology to create a proteomic profile for PD. Over 1300 proteins were measured in serum samples from a de novo Parkinson's (DeNoPa) cohort made up of 85 deep clinically phenotyped drug-naïve de novo PD patients and 93 matched healthy controls (HC) with longitudinal clinical follow-up available of up to 8 years. The analysis identified 73 differentially expressed proteins (DEPs) of which 14 proteins were confirmed as stable potential diagnostic markers using machine learning tools. Among the DEPs identified, eight proteins—ALCAM, contactin 1, CD36, DUS3, NEGR1, Notch1, TrkB, and BTK— significantly correlated with longitudinal clinical scores including motor and non-motor symptom scores, cognitive function and depression scales, indicating potential predictive values for progression in PD among various phenotypes. Known functions of these proteins and their possible relation to the pathophysiology or symptomatology of PD were discussed and presented with a particular emphasis on the potential biological mechanisms involved, such as cell adhesion, axonal guidance and neuroinflammation, and T-cell activation. In conclusion, with the use of advance multiplex proteomic technology, a blood-based protein signature profile was identified from serum samples of a well-characterized PD cohort capable of potentially differentiating PD from HC and predicting clinical disease progression of related motor and non-motor PD symptoms. We thereby highlight the need to validate and further investigate these markers in future prospective cohorts and assess their possible PD-related mechanisms.

Abbreviations: PD, Parkinson's disease; CSF, Cerebrospinal fluid; CNS, Central nervous system; GWAS, Genome-wide association studies; GHR, Growth hormone receptor; DeNoPa, De Novo Parkinson's study; HC, Healthy controls; BL, Baseline; FU, Follow-up; H&Y, Hoehn and Yahr stage; MDS UPDRS, Movement Disorder Society Unified PD Rating Scale; NMSS, Non-motor Symptom Scale; MoCA, Montreal Cognitive Assessment test; BDI, Beck Depression Inventory; GDS, Geriatric Depression Scale; Q1, First quartile; Q3, Third quartile; LIMMA, Linear Models for Microarray data; DEPs, Differentially expressed proteins; CRAN, Comprehensive R Archive Network; LEDD, Levodopa equivalent daily dose; CD209, Cluster of differentiation 209 (C-type lectin domain family 4 member L); FDR, False discovery rate; ALCAM, Activated Leukocyte Cell Adhesion Molecule; TrkB, Tropomyosin receptor kinase B; NOTCH1, Neurogenic locus notch homolog protein 1; NEGR1, Neuronal Growth Regulator 1; CNTN1, Contactin 1; BTK, Bruton's tyrosine kinase; DUS3, Dual Specificity Phosphatase 3; CD36, Cluster of differentiation 36 (Platelet glycoprotein 4); CSK, C-Terminal Src Kinase; LRRK2, Leucine-rich repeat kinase 2; NCAM-L1, Neural cell adhesion molecule L1; CD166, Cluster of differentiation 166; GPI, Glycosylphosphatidylinositol; MAPK, Mitogen-activated protein kinase; NLRP3, NOD-, LRR- and pyrin domain-containing protein 3; BSP, Bone sialoprotein; C1r, Complement 1 subcomponent r; SRCN1, SRC kinase signaling inhibitor 1.

* Corresponding authors.
E-mail addresses: iyahya@hbku.edu.qa (I.Y. Abdi), michael.bartl@med.uni-goettingen.de (M. Bartl), mohammed.dakna@med.uni-goettingen.de (M. Dakna), habdesselem@hbku.edu.qa (H. Abdesselem), n.majbour@ucl.ac.uk (N. Majbour), ctrenkwalder@gmx.de (C. Trenkwalder), oelagnaf@hbku.edu.qa (O. El-Agnaf), brit.mollenhauer@med.uni-goettingen.de (B. Mollenhauer).

[1] These authors contributed equally to this work.
[2] These authors also contributed equally to this work.

https://doi.org/10.1016/j.nbd.2023.105997
Received 25 November 2022; Received in revised form 4 January 2023; Accepted 8 January 2023
Available online 10 January 2023

1. Background

Parkinson's disease (PD), the second most common neurodegenerative disease, is characterized by progressive motor and non-motor symptoms caused by spreading of aggregated α-synuclein pathology (Dickson et al., 2009; Braak and Del Tredici, 2008).

Current PD diagnosis relies on clinical examination. However, there is strong evidence indicating a long, typically non-motor prodromal phase that precedes the manifestation of motor symptoms by several years (Schrag et al., 2015). Symptoms usually arise after the loss of >50% of midbrain dopaminergic neurons (Dickson et al., 2009). The lack of robust early diagnostic and progression biomarkers has markedly hampered the development of a protective treatment strategy, as currently, only symptomatic approaches are available. With the increasing prevalence of PD, likely to double in the near future, there is a growing need for biomarkers that trace the early onset and progression of the disease pathology. Such biomarkers could also facilitate the assessment of treatment efficacy and possibly help with patient stratification.

Over the past decade, efforts have been made to discover and validate biomarkers, in particular, several potential biomarkers have been assessed in cerebrospinal fluid (CSF) as they are the most proximal to the central nervous system (CNS), as well as in blood. These potential biomarkers are currently under investigation, such as α-synuclein species, neurofilament, and lysosomal enzymes (Parnetti et al., 2014; Parnetti et al., 2019; Mollenhauer et al., 2019a; Majbour et al., 2020; Majbour et al., 2021; Oosterveld et al., 2020). None of these, mainly hypothesis-driven markers, have robustly (i.e., validated independently) shown a clear biomarker pattern in PD with a correlation to clinical measures and clinical progression phenotypes, although a distinct biochemical signature is hypothesized to be present as the disease pathology lasts 20-30 years (Fayyad et al., 2019; Oliveira et al., 2021). In addition, increasing evidence points towards a peripheral initiation of the pathology at least in some cases (Borghammer and Van Den Berge, 2019) allowing the detection of a biomarker pattern in peripheral fluids, such as blood. Recent technological advances have widened the avenue for multiplex and large-scale -omics profiling approaches to biomarker discovery. Given the molecular complexity of PD and the fact that only 5–10% of PD cases are caused by monogenic mutations of autosomal dominant genes, the multiplex screening for a panel of markers could provide a more in-depth and valuable insight into the disease pathophysiology and possible molecular phenotypes, allowing for an earlier diagnosis and specific treatment optimization beyond symptomatic therapy (Cherian and Divya, 2020).

Several studies have investigated -omics of PD, like transcriptomics, metabolomics, epigenomics, and proteomics (Lewitt et al., 2017; Posavi et al., 2019; Kia et al., 2021; Hu et al., 2020) with many including genome-wide association studies (GWAS). However, their use leads to cohort-specific results, often lacking deep clinical phenotyping due to the polygenic causes of sporadic PD and are therefore not always reproducible and generalizable.

This study aimed to use a modern aptamer-based platform (Gold et al., 2010) to perform high-quality proteomic profiling (1305 proteins) of cross-sectional serum samples collected at baseline from an established single-center cohort of PD patients and matched healthy controls well-characterized by in-depth clinical phenotyping and data from 8 years of clinical follow-up. In addition, we correlated the multiplex data with longitudinal motor and non-motor scores to elucidate their predictive potential. In addition to creating a proteomic PD profile, the associations of these proteins to α-synuclein serum levels were also investigated to assess possible links between them.

To the best of our knowledge, there have been only two studies that have used the same proteomic analysis approach in a PD cohort. Posavi *et.al* conducted a multicohort longitudinal study on plasma samples that identified four markers that distinguished PD patients from non-PD patients one of which, GHR, was capable of predicting a faster rate of cognitive decline (Posavi et al., 2019). A more recent study by Winchester and colleagues (Winchester et al., 2021) was a more comprehensive multi-cohort discovery and validation study on blood, CSF, and brain tissue that in part validated results found by Posavi et al. Our current study aims to serve as a possible validation of the results reported by these two studies in a high-quality single-center design. The study design and workflow are illustrated in Fig. 1.

2. Methods

2.1. Study participants and cohort information

The De Novo Parkinson's disease (DeNoPa) study is an ongoing prospective longitudinal, single-center cohort study, the demographics, inclusion and exclusion criteria of which have been described in detail previously (Mollenhauer et al., 2013; Mollenhauer et al., 2019b). Recruitment criteria included patients aged between 40 and 85 years, newly diagnosed with PD (de novo) with levodopa exposure for less than two weeks and a minimum of four weeks prior to recruitment. The diagnosis was conducted according to the UK Brain Bank Society criteria (at least two criteria of the following: resting tremor, bradykinesia, and rigidity) (Hughes et al., 1992). Healthy controls (HC) were age-, sex- and education- matched with no family history of PD and showing no pathological condition of the central nervous system. Biannual longitudinal clinical data were collected at baseline (BL), biannually till the maximum 8 years of follow-up (FU) in 104 PD and 94 HC. In this study, baseline (BL) samples from 93 PD and 85 HC were analyzed. With regular biannual clinical FU of the cohort, the 10th year of FU is ongoing enabling a better diagnostic accuracy of participants. We excluded individuals with another neurological disease during FU as published (Mollenhauer et al., 2016).

2.2. Clinical and other available molecular assessments

Clinical assessments at baseline (BL) and follow-up (FU) have been well-described in previous studies (Mollenhauer et al., 2013; Mollenhauer et al., 2016). Assessments included in this study were motor function scores: Hoehn and Yahr stage (H&Y), the Movement Disorder Society Unified PD Rating Scale (MDS UPDRS) part III and the total score; and non-motor functional assessments: Non-Motor Symptom Scale (NMSS), and Montreal Cognitive Assessment Test (MoCA), and the assessment of depression by the self-reported Beck Depression Inventory (BDI), and the Geriatric Depression Scale (GDS) (results are shown in Table 1).

2.3. Protein quantification and preprocessing

The SOMAscan assay platform (SomaLogic, Boulder, USA) was applied for the quantification of 1305 proteins (1.3 K assay) in serum samples and the assay was conducted at the proteomics core of Weill Cornell Medicine-Qatar. Assay details and step-by-step methodology have been described and can be found in the technical white paper by SomaLogic (Gold et al., 2010; Kraemer et al., 2011).

Briefly, serum samples were incubated with chemically modified nucleotides, SOMAmers, that were labelled with fluorophore, photo cleavable linker, and biotin used to immobilized formed complexes on streptavidin coated beads. Unbound proteins were washed away and the SOMAmer bound proteins were biotinylated. All SOMAmer-protein complexes were cleaved from the first set of streptavidin coated beads using UV and the biotinylated SOMAmer bound proteins were recaptured on a second set of streptavidin coated beads. Additional washing steps were done to remove unbound reagents. The bound SOMAmers were released from the beads by the use of denaturing buffer and subsequently hybridized to complementary oligonucleotide in microarray chips and quantified by fluorescence. Serum samples were tested randomly across four plates, each with a set of calibrators and quality control samples run in duplicates for all plates in a blind study format. The hybridization assay data was corrected for systematic biases following the optimized data preprocessing, the methods have been described previously (Candia et al., 2017; SomaLogic, 2015). For data preprocessing, the samples were normalized across plates to adjust for sample-sample variability due to hybridization, using hybridization control samples. This was performed independently for each sample wherein each sample's Relative Fluorescence Unit (RFU) was multiplied by a scale factor. Further, the data was median normalized to correct for intraplate variations due to technical variabilities such as pipetting or differences in overall protein concentration. The grand median across the plates was estimated from the data and each plate-specific median was subtracted from the grand mean. Scale factors for the described normalization processes were expected to be within the accepted range of 0.4-2.5 and all samples' scale factors fell within range. Finally, the data were calibrated to eliminate variations between runs. All calibrators and quality control coefficients of variation were below the exclusion criteria of 0.2.

2.4. Quantification of α-synuclein levels

In addition to the clinical assessments, total serum α-synuclein levels of the samples were quantified for this study. Measurements were conducted using an in-house immunoassay, details of which have been previously published (Majbour et al., 2016).

Serum samples were collected by venipuncture in the morning under fasting conditions and stored in Monovette tubes (Sarstedt, Nümbrecht, Germany). Tubes were centrifuged at 2500*g* at room temperature (20 °C) for 10 min and aliquoted and frozen within 30 min after collection at −80 °C until analysis (Mollenhauer et al., 2013). Measurement of the absorbance of hemoglobin at 414 nm using a NanoDrop™ 2000/2000c spectrophotometer (Thermo Scientific), averaging two technical replicates per sample. Cut-off for hemolysis was set at 0.25 absorbance and samples that were above were excluded from downstream analyses.

2.5. Statistical analysis

All statistical analyses conducted in this study were performed using the R software package (version 4.1.0).

Baseline continuous variables were expressed as mean (standard deviation), median, the interquartile range as given by first and third quartiles Q1, Q3, and the range as given by the min and max values. Group comparisons between PD and HC were performed using the non-parametric Wilcoxon-Test as some of the parameters had non-normal distributions. For the binary variable "sex", the count in each category is provided and the Fischer exact test was used for comparison.

Despite the normalization and quality control methods employed, a technical driver of variability was identified that could not be adjusted for due to its probable orthogonal nature to the outcome variable. To overcome this limitation, conservative statistical parameters (p-value <0.01) were used, and the Boruta algorithm (see below methodology under "Feature selection analysis") as an additional feature selection method was included to ensure the most stable and statistically differential proteins could be identified.

2.6. Differential expression analysis

Normalized SOMAscan data were read using the R *readat* package (Cotton and Graumann, 2016) and the log2 transformed data were used for all subsequent analyses. The *limma* (Linear Models for Microarray Data) R package, version 3.48 (Posavi et al., 2019; Kia et al., 2021; Mollenhauer et al., 2013) was used to compare the expression profiles of PD and HC. Proteins with an adjusted p-value <0.01 and a fold change >1.5 were identified as significantly differentially expressed proteins (DEPs) and selected for further analysis.

2.7. Feature selection analysis

The Boruta algorithm from the CRAN package Boruta is a wrapper built around the random forest classification algorithm. It tries to

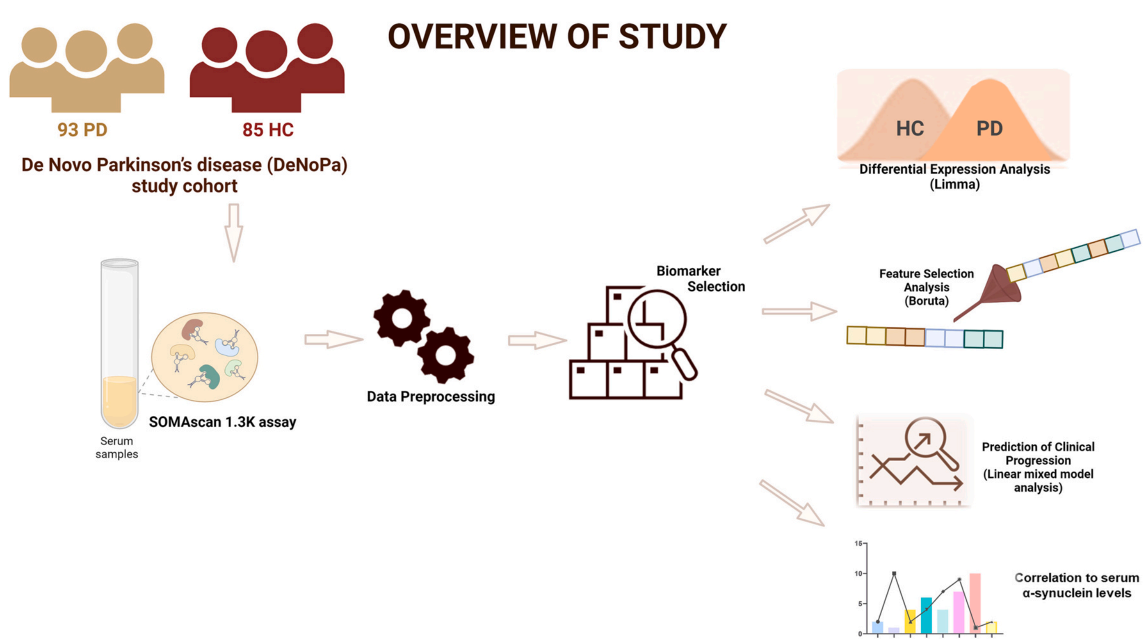

Fig. 1. Overview of study. Serum samples from the De Novo Parkinson's study cohort were analyzed using the SOMAscan 1.3 K assay kit from SomaLogic. Bioinformatic analyses: differential expression analysis, feature selection analysis, and linear mixed model analyses were used to identify a statistically significant panel of diagnostic and predictive biomarkers for Parkinson's disease. (Created with BioRender.com).

Table 1
Demographic data of participants with PD and healthy controls and clinical data assessments.

DEMOGRAPHICS	HC (*N* = 93)	PD (*N* = 85)	*p*-value
SEX			0.581
-Female	41 (44.1%)	34 (40.0%)	
-Male	52 (55.9%)	51 (60.0%)	
-Missing	0	0	
AGE at Baseline (y)			0.836
- Mean (SD)	64.62 (6.95)	64.48 (9.97)	
- Median (Q1, Q3)	64 (60, 70)	66. (57.00, 72.00)	
- Min - Max	49.00 - 84.00	40.00 - 84.00	
- Missing	0	0	
BMI			**0.03**
- Mean (SD)	26.46 (4.43)	27.95 (4.93)	
- Median (Q1, Q3)	25.62 (23.11, 28.86)	27.34 (24.39, 29.98)	
- Min - Max	19.27 - 42.25	20.03 - 44.79	
- Missing	0	0	
CLINICAL DATA ASSESSMENTS			
	HC (N = 93)	**PD (N = 85)**	**p-value**
MOTOR FUNCTION SCORES			
MDS UPDRS part III			**< 0.001**
- Mean (SD)	0.68 (1.47)	22.05 (11.14)	
- Median (Q1, Q3)	0.00 (0.00, 0.00)	21.00 (13.00, 29.00)	
- Min – Max	0.00 - 6.00	3.00 - 54.00	
- Missing	0	0	
MDS UPDRS total score			**< 0.001**
- Mean (SD)	3.57 (3.39)	36.02 (17.74)	
- Median (Q1, Q3)	3.00 (1.00, 5.00)	34.00 (23.00, 45.00)	
- Min – Max	0.00 - 15.00	7.000 - 84.00	
- Missing	0	0	
Hoehn & Yahr stage			**< 0.001**
- Mean (SD)	0.00 (0.00)	1.97 (0.68)	
- Median (Q1, Q3)	0.00 (0.00, 0.00)	2.00 (2.00, 2.00)	
- Min – Max	0.00 - 0.00	1.00 - 3.00	
- Missing	0	0	
COGNITIVE FUNCTION SCORES			
MoCA			**0.035**
- Mean (SD)	26.34 (2.06)	24.98 (3.31)	
- Median (Q1, Q3)	26.00 (25.00, 28.00)	26.00 (23.00, 27.75)	
- Min - Max	22.00 - 30.00	16.00 - 30.00	
- Missing	11	31	
NON-MOTOR FUNCTION SCORES			
Non-Motor Symptoms Scale (NMSS)			**0.035**
- Mean (SD)	3.91 (2.64)	7.44 (4.15)	
- Median (Q1, Q3)	3.00 (2.00, 5.00)	7.00 (4.00, 10.00)	
- Min - Max	0.00 - 12.00	0.00 - 19.00	
- Missing	0	0	
COGNITIVE FUNCTION SCORES			
MoCA			**0.035**
- Mean (SD)	26.34 (2.06)	24.98 (3.31)	
- Median (Q1, Q3)	26.00 (25.00, 28.00)	26.00 (23.00, 27.75)	
- Min - Max	22.00 - 30.00	16.00 - 30.00	
- Missing	11	31	
DEPRESSION RATING SCORES			
BDI			**<0.001**
- Mean (SD)	3.73 (4.91)	8.19 (6.43)	
- Median (Q1, Q3)	2.00 (1.00, 5.00)	7.00 (3.24, 12.00)	
- Min - Max	0.00 - 30.00	0.00 - 28.00	
- Missing	0	1	
GDS			**<0.001**
- Mean (SD)	1.27 (1.99)	3.32 (2.87)	
- Median (Q1, Q3)	1.00 (0.00, 2.00)	3.00 (1.00, 5.00)	
- Min – Max	0.00 - 13.00	0.00 - 12.00	
- Missing	3	4	

Abbreviations: HC, healthy control; PD, Parkinson's disease; BMI, body mass index; MDS UPDRS, Movement Disorder Society Unified Parkinson's Disease Rating Scale; SD, standard deviation; Q1, first quartile; Q3, third quartile, MCI: Mild cognitive impairment; MoCA, Montreal Cognitive Assessment; BDI, Beck Depression Inventory; GDS, Geriatric Depression Scale; MADRS, Montgomery-Åsberg Depression Rating Scale.

capture all the important and interesting features in the dataset with respect to outcome variables of PD versus HC. Firstly, the algorithm adds randomness to the dataset by creating shuffled copies of all features which are called Shadow Features. Then, it trains a random forest classifier on this extended dataset (original attributes plus shadow attributes), applies a feature importance measure (The Mean Decrease Accuracy), and evaluates the importance of each feature. At every iteration, the Boruta algorithm checks whether a real feature is more important than the other. It then removes features that are deemed highly unimportant. As a stopping rule, we used a maximum of 500 random forest as indicated by the parameter maxRuns in the Boruta function. With actual CRAN implementation of Boruta, where warm-up rounds are removed, and the multiple testing corrections are introduced, it should find all features that are either strongly or weakly relevant to the PD diagnosis. In our dataset, where a complex covariance structure between the SOMAmers is expected, such an algorithm that takes into account the multi-variable relationships and the interactions between them is much more suitable than the linear models of limma.

2.8. Linear mixed-effects model analysis

For the continuous clinical scores, longitudinal modeling was undertaken via a random slope linear mixed model using the R package lme4 for all dependent variables. The distribution of most clinical scores was right-skewed. We, therefore, transformed the dependent variables to the square root scale to achieve normal distributions of the residual errors in the models. For all models, we checked this assumption by checking the q-q plots of the residual errors. All models were adjusted for age, sex, and levodopa equivalent daily dose (LEDD). The clinical FU period considered was six years. Furthermore, the age was centered around the mean to allow simple interpretations of the intercepts in the models. The SOMAmers were transformed to log2 and added to the models as a baseline covariate to assess its predictive association with the longitudinal expression of the clinical outcome.

2.9. Pathway clustering

We performed an advanced pathway analysis and built a cluster, based on a systematic literature search of MedLine for the detected significantly differently expressed markers, correlated markers, and the search terms "Parkinson's disease" and "Neurodegeneration". According to this pathway analysis, we performed a clustering into inflammation groups, characterized by direct involvement of immunological processes, including the innate as well as the adaptive immune systems, suggesting a direct involvement of immunological processes, like CD209 antigen and cell biology, including the aspects cell: signaling/ growth/ proliferation/ metabolism/ adhesion/ vesicular transport (Durafourt et al., 2012).

3. Results

3.1. Differential expression analysis and feature selection analysis

Of the 1305 measured proteins, 73 were identified to be differentially expressed between PD patients and HC. Of these proteins, 46 were downregulated and 27 were upregulated in PD (Fig. 2 a, b, Table 2). Table 2 reports the top 15 upregulated, and downregulated DEPs assessed with *limma* analysis, including protein name and gene ID as well as the adjusted *p*-value and FDR.

To identify the most robust markers, feature selection analysis *Boruta* was conducted on the complete proteomic profile of 1305 markers and

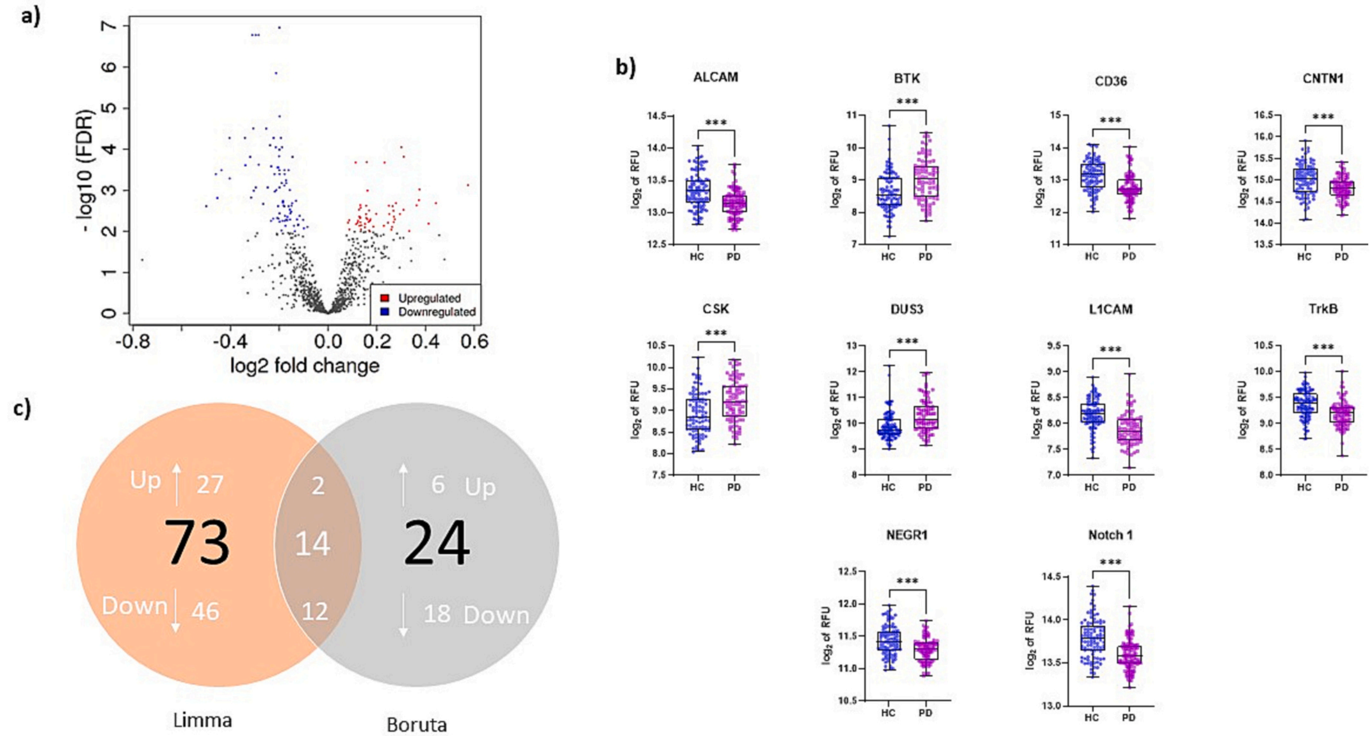

Fig. 2. Differential expression analysis. a) Volcano plot of differentially expressed proteins between PD and HC ($p < 0.01$); b) Boxplot showing the distribution of the 10 proteins' levels in PD and HC which correlated with PD disease progression scores. c) Venn diagram of the overlap of identified DEPs between PD and HC.

24 differentially expressed markers were identified. Of these identified 24 markers, 14 overlap with those identified using the *limma* analysis thereby confirming that they are the most stable markers capable of differentiating PD samples from HC samples. (Fig. 1c and Table 3).

3.2. Quantification of α-synuclein levels

Serum α-synuclein levels were significantly increased in patients with PD compared to controls (Mean values: 21 ng/ml (17.8-25.1) vs 14.5 ng/ml (11.8-17.1); *p*-values<0.001) (Supplementary Fig. 1). Hemoglobin levels were also measured, and only one sample was above the set cut-off for hemolysis and therefore excluded from the analysis. Despite having low hemoglobin levels, two samples (1 PD and 1HC) were measured to be outliers with high levels of α-synuclein. Exclusion of these outliers did not affect the significance of the results observed (Mean values: PD- 19.6 ng/ml (17.8-21.3) vs HC- 13.6 ng/ml (12.2-15.1); p-values<0.001) (Supplementary Fig. 1).

3.3. Correlations with clinical and molecular assessment scores

To identify any associations between longitudinal non-motor and motor functional scores and the identified biomarker levels measured at BL, a linear mixed-effects model was fitted including age, sex, disease duration, PD diagnosis, and log 10 of levodopa equivalented daily dose (LEDD) as fixed effects (Tomlinson et al., 2010).

Results of the linear mixed-effects model analysis identified eight markers- Notch1, ALCAM, DUS3, CD36, Contactin 1, TrkB, NEGR1, and BTK- that significantly correlated with changes in non-motor and PD progression scores over 8 years. Four of the eight markers- Notch1, ALCAM, DUS3, and CD36- were also identified as potential diagnostic markers via *limma* and *Boruta* analyses.

Motor progression measured by MDS UPDRS total score significantly correlated with the proteins ALCAM and TrkB, which were found to be downregulated in PD (*p*-value = 0.03). The cognitive decline assessment score (MoCA) significantly correlated with the levels of the cell adhesion proteins Notch 1, NEGR1, and CNTN1 (*p*-values = 0.01, 0.02, and 0.01, respectively) all of which were found to be downregulated in PD.

The proteins BTK and DUS3 were upregulated in PD, and both negatively correlated with the depression rating scale (BDI; *p*-values = 0.02 and 0.01, respectively), implicating more depressive symptoms in those with lower serum levels. ALCAM, along with another downregulated protein, CD36, both correlated positively with the depression rating scale (GDS; p-values = 0.03 and 0.04, respectively).

Baseline serum total α-synuclein levels were tested for their correlation to the identified 14 DEPs, which were differentially expressed in both methods (*limma* and *Boruta*). No overlap in the significantly correlated markers was observed between the correlation to serum α-synuclein levels. By including all the 73 differently expressed marker proteins in the analysis, the tyrosine protein kinases BTK and CSK showed a significant correlation to serum t-α-synuclein levels (p-value = 0.03). Both proteins were significantly upregulated in PD. A summary of the linear mixed-effects model analysis results mentioned above was collected in Table 4.

3.4. Pathway clustering

Extended pathway analysis revealed that the 14 differentially expressed markers that were identified had a direct connection to inflammatory processes and neurodegeneration and PD. Furthermore, we found that Notch 1 belonged to the complex of cell biology “signaling”, NCAM-L1, ALCAM, TrkB, and Siglec-7 connected to cell biology “adhesion”, ASGR1, NCAM-L1 to cell biology “vesicular transport”, TrkB, CNTN1, and NEGR1 to cell biology “growth” and cell biology “proliferation”, and finally DUS3 to the cell biology complex “metabolism”. Alterations in molecules involved in cell adhesion, cell-cell interaction, and processes related to the extracellular matrixes were most common (Table 5).

Table 2
List of top 15 ranked differentially upregulated and downregulated proteins between HC and PD using Limma.

Top Downregulated Proteins				
	Target Full Name	Target	logFC	adj.P. Val
1	Neurogenic locus Notch homolog protein 1	Notch 1	-0.87	<0.001
2	Fructose-bisphosphate aldolase A	aldolase A	-0.85	<0.001
3	Neural cell adhesion molecule L1	NCAM-L1	-0.84	<0.001
4	Complement component C1q receptor	C1QR1	-0.84	<0.001
5	CD166 antigen	ALCAM	-0.78	<0.001
6	Prothrombin	Prothrombin	-0.72	<0.001
7	Lumican	Lumican	-0.72	<0.001
8	C-type mannose receptor 2	MRC2	-0.72	<0.001
9	Cadherin-5	Cadherin-5	-0.71	<0.001
10	BDNF/NT-3 growth factors receptor	TrkB	-0.7	<0.001
11	Cell adhesion molecule-related/ downregulated by oncogenes	CDON	-0.69	<0.001
12	dCTP pyrophosphatase 1	XTP3A	-0.67	<0.001
13	Platelet glycoprotein 4	CD36 ANTIGEN	-0.67	<0.001
14	Complement decay-accelerating factor	DAF	-0.67	<0.001
15	Contactin-4	Contactin-4	-0.66	<0.001

Top Upregulated Proteins				
	Target Full Name	Target	logFC	adj.P. Val
1	Dual-specificity tyrosine-phosphorylation-regulated kinase 3	DYRK3	0.75	<0.001
2	Tumor necrosis factor ligand superfamily member 14	LIGHT	0.73	<0.001
3	Ephrin-B1	EFNB1	0.72	<0.001
4	Cathepsin G	Cathepsin G	0.7	<0.001
5	3-phosphoinositide-dependent protein kinase 1	PDPK1	0.68	<0.001
6	Asialoglycoprotein receptor 1	ASGR1	0.64	<0.001
7	'Inosine-5''-monophosphate dehydrogenase 2'	IMDH2	0.62	<0.001
8	Dual-specificity protein phosphatase 3	DUS3	0.61	<0.001
9	Ephrin type-B receptor 4	EphB4	0.61	<0.001
10	Breast cancer anti-estrogen resistance protein 3	BCAR3	0.61	<0.001
11	Cellular tumor antigen p53	p53	0.6	<0.001
12	Glutathione S-transferase P	GSTP1	0.6	<0.001
13	'cGMP-dependent 3″,5″-cyclic phosphodiesterase'	cGMP-stimulated PDE	0.6	<0.001
14	Growth factor receptor-bound protein 2	GRB2 adapter protein	0.6	<0.001
15	Abelson tyrosine protein kinase 2	ABL2	0.59	<0.001

4. Discussion

Biomarkers are an essential basis for early and reliable diagnosis of molecular complex PD, including the differentiation from other neurodegenerative diseases, the assessment of disease progression, as well as for drug development as causative treatment to stop or at least slow disease progression. With more than six million affected patients worldwide and a fast-rising prevalence, there is an urgent need to assess reliable biomarkers for state, rate, and fate in PD (Feigin et al., 2019). However, biomarkers must meet various criteria to be broadly applicable and reliable. CSF biomarkers, despite their proximity to the disease processes taking place in the CNS, have largely failed in the past and are not accessible without invasive methods e.g., lumbar puncture (Fayyad et al., 2019; Oliveira et al., 2021; Poewe et al., 2017). Whereas, Blood-based biomarkers are preferred to CSF biomarkers and with the increasing evidence for peripheral PD pathology, there has been a raised opportunity for the discovery of peripheral PD biomarkers. As the most important effector molecules in the body, proteins can be potentially used as predictors, disease-modifying targets, and surrogate markers of disease progression in advanced PD research. (Williams-Gray et al., 2016).

Based on these considerations, this study used a large-scale proteomics approach to identify differentially expressed proteins (DEPs) between deeply phenotyped de novo PD patients and high-quality matched HC cross-sectionally, with the ultimate aim of identifying potential diagnostic markers for PD. All the identified DEPs, were used to conduct predictive modeling to test the ability of the marker levels at baseline to predict clinical disease progression and the related motor and non-motor symptoms up to 8 years after diagnosis. Results generated from the analyses conducted were divided into; 1) potential diagnostic markers capable of discriminating between PD and HC and constitute fourteen proteins confirmed by both the *Boruta* and *limma* tools; 2) nine potentially predictive markers for clinical symptoms and progression of PD as these proteins were found to be significantly correlated to established clinical scores. Among the proteins identified as potentially diagnostic or predictive markers, four proteins overlapped as both potential diagnostic and predictive markers for PD and its clinical symptoms and progression.

This study also included a further analysis investigating possible associations between baseline total α-synuclein levels and the identified DEPs levels. Growing evidence has shown α-synuclein to be a key protein in PD pathology with several studies showing that despite its ubiquitous expression throughout the body, the presence of disease-associated α-synuclein is not only limited to the brain but is also present in the periphery, even in early or prodromal stages of PD (Williams-Gray et al., 2016). Several studies have shown higher levels of α-synuclein in plasma of PD compared to controls which was in agreement with results found in our study (Chang et al., 2020; Zubelzu et al., 2022; Syafrita and Susanti, 2022). There are studies that have reported

Table 3
List of overlapping differentially up-regulated and down-regulated proteins between Limma and Boruta analyses.

	Target full name	Target	UniProt	Entrez gene ID	Entrez gene symbol	Direction of expression
1	Lumican	Lumican	P51884	4060	LUM	Down
2	Complement component C1q receptor	C1QR1	Q9NPY3	22,918	CD93	Down
3	Sialic acid-binding Ig-like lectin 7	Siglec-7	Q9Y286	27,036	SIGLEC7	Down
4	Midkine	Midkine	P21741	4192	MDK	Down
5	Platelet glycoprotein 4	CD36 ANTIGEN	P16671	948	CD36	Down
6	Transforming growth factor beta receptor type 3	TGF-b R III	Q03167	7049	TGFBR3	Down
7	CD209 antigen	DC-SIGN	Q9NNX6	30,835	CD209	Down
8	Dual-specificity protein phosphatase 3	DUS3	P51452	1845	DUSP3	Up
9	Neural cell adhesion molecule L1	NCAM-L1	P32004	3897	L1CAM	Down
10	dCTP pyrophosphatase 1	XTP3A	Q9H773	79,077	DCTPP1	Down
11	Neurogenic locus Notch homolog protein 1	Notch 1	P46531	4851	NOTCH1	Down
12	CD166 antigen	ALCAM	Q13740	214	ALCAM	Down
13	Asialoglycoprotein receptor 1	ASGR1	P07306	432	ASGR1	Up
14	Fructose-bisphosphate aldolase A	aldolase A	P04075	226	ALDOA	Down

Table 4
Results of mixed-effects linear models.

	Estimate	Std. Error	p-value
MOVEMENT DISORDER SOCIETY UNIFIED PD RATING SCALE (MDS UPDRS) TOTAL SCORE			
CD166 antigen (ALCAM)			
(Intercept)	-7.79	4.41	0.08
AGE	0.05	0.01	0.00
SEX: male	0.09	0.16	0.59
DIAGNOSIS: PD	4.43	0.21	0.00
TIME	0.16	0.03	0.00
log10(LEDD)	0.10	0.07	0.11
CD166 antigen	0.71	0.33	**0.03**
BDNF/NT-3 growth factor receptor (TrkB)			
(Intercept)	-4.97	2.82	0.08
AGE	0.05	0.01	0.00
SEX: male	0.02	0.16	0.90
DIAGNOSIS: PD	4.42	0.20	0.00
TIME	0.15	0.03	0.00
log10(LEDD)	0.10	0.07	0.11
BDNF/NT-3 growth factors receptor	0.71	0.30	**0.02**
NON-MOTOR SYMPTOM SCORE (NMSS)			
Dual-specificity protein phosphatase 3 (DUS3)			
(Intercept)	4.23	0.87	0.00
AGE	0.02	0.01	0.01
SEX: male	-0.30	0.10	0.00
DIAGNOSIS: PD	0.80	0.12	0.00
TIME	0.00	0.01	0.96
log10 (LEDD)	0.07	0.03	0.01
Contactin-1	-0.28	0.09	**0.01**
MONTREAL COGNITIVE ASSESSMENT TEST (MoCA)			
Contactin-1 (CNTN1)			
(Intercept)	2.80	0.86	0.00
AGE	-0.01	0.00	0.00
SEX: male	-0.02	0.04	0.59
DIAGNOSIS: PD	-0.08	0.05	0.07
TIME	0.02	0.00	0.00
log10 (LEDD)	-0.04	0.01	0.00
Contactin-1	0.15	0.06	**0.01**
Neurogenic locus Notch homolog protein 1 (Notch 1)			
(Intercept)	1.87	1.22	0.13
AGE	-0.01	0.00	0.00
SEX: male	-0.03	0.04	0.33
DIAGNOSIS: PD	-0.07	0.05	0.15
TIME	0.02	0.00	0.00
log10 (LEDD)	-0.04	0.01	0.00
Neurogenic locus Notch homolog protein 1	0.23	0.09	**0.01**
Neuronal growth regulator 1 (NEGR1)			
(Intercept)	2.67	0.99	0.01
AGE	-0.01	0.00	0.00
SEX: male	-0.04	0.04	0.28
DIAGNOSIS: PD	-0.08	0.05	0.08
TIME	0.02	0.00	0.00
log10 (LEDD)	-0.05	0.01	0.00
Neuronal growth regulator 1	0.21	0.09	**0.02**
BECK DEPRESSION INVENTORY SCORE (BDI)			
Tyrosine protein kinase (BTK)			
(Intercept)	4.14	1.05	0
AGE	0.02	0.01	0.01
SEX: male	-0.31	0.15	0.04
DIAGNOSIS: PD	1.11	0.17	0
TIME	-0.02	0.02	0.34
log10(LEDD)	-0.03	0.04	0.53
Tyrosine protein kinase BTK	-0.27	0.12	0.02
Dual-specificity protein phosphatase 3 (DUS3)			
(Intercept)	5.28	1.28	0.00
AGE	0.02	0.01	0.01
SEX: male	-0.32	0.15	0.04
DIAGNOSIS: PD	1.14	0.17	0.00
TIME	-0.02	0.02	0.35
log10 (LEDD)	-0.03	0.04	0.53
Dual-specificity protein phosphatase 3	-0.35	0.13	**0.01**
GERIATRIC DEPRESSION SCALE (GDS)			
Platelet glycoprotein (CD36)			
(Intercept)	-2.94	1.89	0.12
AGE	0.02	0.01	0.02
SEX: male	0.08	0.13	0.51
DIAGNOSIS: PD	0.89	0.14	0.00
TIME	-0.02	0.01	0.09
log10 (LEDD)	-0.04	0.03	0.26
Platelet glycoprotein 4	0.28	0.14	**0.05**
CD166 antigen (ALCAM)			
(Intercept)	-6.26	3.22	0.05
AGE	0.02	0.01	0.03
SEX: male	0.04	0.12	0.74
DIAGNOSIS: PD	0.89	0.14	0.00
TIME	-0.02	0.01	0.11
log10(LEDD)	-0.04	0.03	0.26
CD166 antigen	0.53	0.24	**0.03**
SERUM TOTAL-α-SYNUCLEIN			
Tyrosine protein kinase (CSK)			
(Intercept)	-1.21	2.24	0.59
AGE	0.00	0.01	0.86
SEX: male	-0.05	0.24	0.82
DIAGNOSIS: PD	0.50	0.28	0.08
TIME	0.06	0.07	0.40
log10 (LEDD)	-0.17	0.13	0.18
Tyrosine protein kinase CSK	0.55	0.25	**0.03**
Tyrosine protein kinase (BTK)			
(Intercept)	0.18	1.69	0.92
AGE	0.00	0.01	0.99
SEX: male	0.04	0.24	0.88
DIAGNOSIS: PD	0.51	0.28	0.07
TIME	0.06	0.07	0.42
log10 (LEDD)	-0.16	0.13	0.21
Tyrosine protein kinase BTK	0.41	0.19	**0.04**

Linear mixed model analysis of differentially expressed proteins to change in clinical scores related to motor and non-motor symptoms in Parkinson's disease over a period of 8 years.

discrepant results however these could be attributed to possible blood contamination or hemolysis confounding the quantification of α-synuclein as red blood cells store high levels of the protein (Barbour et al., 2008; Albillos et al., 2021). Measurements of hemoglobin levels in the samples tested in this study were taken into consideration to avoid this confounding effect.

Using extensive literature review, the functions of the different proteins identified and possible explanations for their associations to the PD pathology and symptoms were investigated (Table 5). Henceforth discussed are the; 1) four proteins identified as both potential diagnostic and predictive markers; 2) two proteins that significantly correlated with total-α-synuclein; 3) finally two other proteins that have been reported to be linked to α-synuclein and PD pathology and in our study was differentially expressed between HC and PD however did not reach statistical power to be classified as a diagnostic or predictive marker for PD or its symptoms.

The four proteins classified as both potential diagnostics and predictive markers include Notch 1, ALCAM, DUS3, and CD36. Notch 1, a transmembrane protein that plays a critical role in neuronal and glial

Table 5
Biomarker Information and pathway clustering.

Protein	Uniprot ID	Regulation in PD	Correlation to longitudinal PD motor and non-motor symptoms	Biological function	Evidence in neuronal pathways/Clinical implications	Cluster
Notch 1	P46531	↓	Positive correlation with MoCA Score	Receptor for membrane-bound ligands	Notch signaling gets modulated by LRRK2 through endosomal pathways)(Imai et al., 2015)	Cell biology (**signaling**/ growth/proliferation/ metabolism/adhesion/ vesicular transport)
Aldolase A	P04075	↓	No correlation	Enzyme in Glycolysis/ Gluconeogenesis	Target for Autoantibodies in hyperkinetic movement disorders)(Privitera et al., 2013)	Inflammation
NCAM-L1	P32004	↓	No correlation	Cell adhesion, axon development, neuron projection	Increased α-synuclein in NCAM-L1 immunocaptured exosomes, differentiation against HC, MSA and CBS (Jiang et al., 2021)	Cell biology (signaling/ growth/ proliferation/ metabolism/ **adhesion**/ **vesicular transport**)
C1QR1	Q9NPY3	↓	No correlation	Receptor for C1q, mediation of phagocytosis, elimination and connection of synapses	↓ of C1q in exosomes in PD (Jiang et al., 2019)	Inflammation
ALCAM	Q13740	↓	Positive correlation with MDS UPDRS Total Score and GDS	Cell adhesion molecule	Axonal growth of midbrain dopaminergic neurons (Bye et al., 2019)	Cell biology (signaling/ growth/proliferation/ metabolism/**adhesion**/ vesicular transport)
Lumican	P51884	↓	No correlation	Collagen fibril binding and organization, ECM component	Proteoglycan in the brain, tumor-cell interaction, inflammation, and angiogenesis (Nikitovic et al., 2014)	Inflammation
TrkB	Q16620	↓	Positive correlation with MDS UPDRS Total Score	Various, including response to nerve growth factor/brain-derived neurotrophic factor. Maturation of CNS and PNS	BDNF-associated activation of NRF2 in astrocytes is neuroprotective for dopaminergic neurons via the transfer of antioxidants (Ishii et al., 2019)	Cell biology (signaling/ **growth/proliferation**/ metabolism/**adhesion**/ vesicular transport)
CD36 Antigen	P16671	↓	Positive correlation with GDS	Amyloid-β clearance, apoptotic cell clearance, blood coagulation	Mediation of amyloid-β and α-synuclein driven microglial activation (Su et al., 2008)	Inflammation
Siglec-7	Q9Y286	↓	No correlation	Cell adhesion	Association with amyloid-β plaque degradation (Tortora et al., 2022)	Cell biology (signaling/ growth/proliferation/ metabolism/**adhesion**/ vesicular transport)
Midkine	P211741	↓	No correlation	Heparin-binding growth factor or cytokine involved in T-cell activation/differentiation and cerebral cortex development	Genetic loss of Midkine leads to dopaminergic cell death in the midbrain (Prediger et al., 2011)	Inflammation
TGFBR3	Q03167	↓	No correlation	Various including Intracellular signaling in TGF- β pathways, immune response	TGF deficiency leads to nigrostriatal degeneration, and upregulation protects against MPTP toxicity (Tesseur et al., 2017)	Inflammation
CD209 antigen	Q9NNX6	↓	No correlation	Multiple immune functions including regulation of T-cell differentiation/proliferation	M2-activation status in brain microglia (Durafourt et al., 2012)	Inflammation
CNTN1	Q12860	↓	Positive correlation with MoCA Score	Axon guidance, myelin formation, brain development	CNTN1 is reduced in PD CSF and present in Lewy bodies (Chatterjee et al., 2019, 2020)	Cell biology (signaling/ **growth/proliferation**/ metabolism/adhesion/ vesicular transport)
NEGR1	Q7Z3B1	↓	Positive correlation with MoCA Score	Brain development, locomotory behavior	Controls hippocampal development (Noh et al., 2019)	Cell biology (signaling/ **growth/proliferation**/ metabolism/adhesion/ vesicular transport)
ASGR1	P07306	↑	No correlation	Receptor-mediated endocytosis	Downregulated in intracranial arachnoid cyst tissue (Aarhus et al., 2010)	Cell biology (signaling/ growth/proliferation/ metabolism/adhesion/ **vesicular transport**)
DUS3	Q96G46	↑	Negative correlation with BDI and NMSS	tRNA-synthesis	tRNA-fragments can distinguish between PD and HC in a sex-specific manner (Magee et al., 2019)	Cell biology (signaling/ growth/proliferation/ **metabolism**/adhesion/ vesicular transport)
BTK	Q06187	↑	Negative correlation with BDI	Innate immune response, B-cell activation	Inhibition of BTK can reduce microglial activation, phagocytosis, and neurodegeneration in AD models (Keaney et al., 2019)	Inflammation
CSK	Q06187	↑	Positive correlation with Serum Total α-synuclein	Various including T-cell costimulation, adaptive immune response	Microglial homeostasis with link to PD and AD pathology (Portugal et al., 2021)	Inflammation

Abbreviations: PD: Parkinson's Disease, HC: Healthy Control, MSA: Multiple Systems Atrophy, CBS: Corticobasal Syndrome, CNS: Central Nervous System, AD: Alzheimer's disease.

proliferation (Lütolf et al., 2002), was most significantly downregulated in PD serum samples against HC (Costa et al., 2003). The Notch 1 signaling pathway not only controls neurogenesis but is further involved in axonal growth and guidance mechanisms (Šestan et al., 1999; Ables et al., 2011; Song and Giniger, 2011; Ding et al., 2016). Studies have shown Notch 1 expression to be altered in Alzheimer's disease and PD with possible links to α-synuclein expression (Cho et al., 2019; Crews et al., 2008). Further, cognitive impairment has been associated with decreased Notch 1 levels (Ding et al., 2016). A study by Imai et al. showed links between altered the Notch signaling pathway and PD by mutated LRRK2, one of the most common PD-associated genes (Imai et al., 2015). These studies suggest that dysfunctional neurogenesis and axonal guidance mechanisms are involved in the PD-specific pathology and support our finding that Notch 1 serum levels in PD correlated with the MoCA score, frequently used to screen, and rate cognitive functions in PD.

Altered neural circuitry impaired axonal guidance and neurogenesis, cell growth and proliferation, metabolism and signaling and can potentially cause neurodevelopmental disorders as well as contribute to the progression of neurodegenerative diseases including PD (Canter et al., 2016; Busche and Konnerth, 2016). Proteins also assessed in this study that are associated with roles in axonal guidance and growth mechanisms, include the cell adhesion molecules, ALCAM, and NCAM-L1 (Pollerberg et al., 2013; Singh et al., 2019). ALCAM, also known as CD166 antigen, has a PD-specific connection as it guides the axonal growth of dopaminergic neurons in the midbrain, the main region of PD pathology (Bye et al., 2019). In this study, serum levels of ALCAM were downregulated in PD patients and positively correlated with the MDS UPDRS total score, implicating an association between higher serum levels and worse motor performance. ALCAM is a cell adhesion molecule reported in the literature to function in axonal guidance and synaptic plasticity, and impairments in both mechanisms have been found in patients with major depression disorder (Pollerberg et al., 2013; Singh et al., 2019; Kroenke et al., 2001; Khairova et al., 2009). This is supported by the positive correlation with the GDS depression scale in our study, and by a previous study that showed altered peripheral levels of ALCAM in response to anti-depression treatment (Ma et al., 2019).

Dual-specificity phosphatase-3 (DUS3) belongs to a superfamily of protein tyrosine phosphatases that work on phosphorylating tyrosine, serine, and threonine residues. It was also found to regulate cell differentiation and growth by modulating pathways such as negatively regulating members of the MAPK signaling pathway (Nunes-Xavier et al., 2019). Dysregulation of the signaling pathways such as MAPK has been shown to occur in PD pathogenesis. However, the exact mechanisms involved are yet to be identified and validated (Bohush et al., 2018; Kim and Choi, 2010). In addition to being found as significantly upregulated in PD, DUS3 correlated with the NMSS score, assessing non-motor symptoms as well as the depression scores as measured with the GDS and the BDI. These findings indicate the possible involvement of DUS3 in PD pathology with direct connections to the clinical phenotype and potentially a therapeutic target, worth evaluating in further investigations.

Platelet glycoprotein 4, CD36, is involved in mediating microglial pro-inflammatory response (Su et al., 2008). Levels of CD36 were downregulated in PD and also correlated with the GDS depression score. In another study, CD36 deficiency was found to impact depressive-like behavior by the inflammasome hypothesis of depression (Bai et al., 2021). It has also been reported to work in conjunction with tyrosine kinase Fyn, also among the likely upregulated proteins in our dataset, to regulate the uptake of misfolded α-synuclein via the inflammasome pathway (Panicker et al., 2019).

The tyrosine kinases BTL and CSK are associated with inflammatory processes and correlated with total-α-synuclein levels in serum in our study presenting a possible connection to α-synuclein pathology. CSK, also known as c-Src, is a non-receptor tyrosine kinase of the Src family and is also capable of phosphorylating α-synuclein (Ellis et al., 2001). c-Src has also been reported to be involved in the cell-to-cell transmission of α-synuclein (Choi et al., 2020) and studies have shown that inhibiting c-Src activation attenuated neuroinflammation, α-synuclein propagation, and protected dopaminergic neurons (Choi et al., 2020; Yang et al., 2020), thereby supporting the involvement of these kinases in PD pathology.

BTK is predominantly expressed in B cells and is a known positive regulator of the NLRP3 inflammasome, which has been shown in studies to be activated in the periphery of PD patients and can be triggered by α-synuclein (Pal Singh et al., 2018; Weber, 2021; Ito et al., 2015; Fan et al., 2020). In addition to its correlation to α-synuclein levels in PD, the results of our study also associated it with depression, a common non-motor symptom of PD, that continues to be studied less than other PD symptoms. BTK correlated with changes in the BDI depression rating scale. Given the involvement of BTK in the NLRP3 inflammasome pathway, its link to depression is an interesting finding as the inflammasome has been reported to be a key mediator in mechanisms that regulate psychological stressors including depression. This phenomenon has more recently been studied and the term "inflammasome hypothesis of depression" has been coined (Wong et al., 2016; Sahin and Aricioglu, 2016; Iwata et al., 2013).

Also worth discussing are two markers that displayed potential diagnostic capabilities but did not correlate with clinical scores or total-α-synuclein levels. These markers have been reported to be related to α-synuclein and PD-related pathology including inflammation, a specific feature with increasing evidence suggesting its links to PD pathology (Caggiu et al., 2019). Interestingly, these two markers, Midkine and CD209 antigen, that were significantly under expressed in PD compared to HC in our study, play a role in T-cell regulation and activity (Muramatsul et al., 2011; Jung et al., 2004; Herradõn and Pérez-García, 2014). Studies have shown that Midkine deficiency resulted in dopaminergic system dysfunction as well as olfactory and cognitive impairments, both known symptoms of PD and modulates immune responses/inflammation in neurodegenerative disorders (Prediger et al., 2011; Ohgake et al., 2009). CD209 antigen gets expressed by brain microglia and blood-derived macrophages in the M2-activation status, which is suggested to include repair processes and regulation of inflammation, further it is involved in T-cell activation and differentiation (Durafourt et al., 2012).

Finally, there were markers that did not satisfy statistical tests to be considered diagnostic markers for PD however they correlated with cognitive clinical scores and MDS UPDRS Total score. These markers were cell adhesion molecules, Contactin 1, TrkB and NEGR1 which were all downregulated in our study.

We compared our findings with the results of previous studies, applying SOMAmer technology in PD samples. The protein markers identified by Posavi et al. *(*Posavi et al., *2019)*, BSP, C1r, GHR, and SRCN1 were also identified in this study as differentially expressed (p-value$<$0.05) while using the more stringent feature selection analysis, Boruta, confirmed the differential expression of BSP in PD compared to control samples in our cohort. Correlating our results with the study by Winchester et al. showed an overlap of the top DEP, Notch 1 which was also found to be downregulated in PD (Winchester et al., 2021).

Our study includes the following limitations that have to be addressed. (I) DeNoPa cohort is a single-center cohort well characterized with stringent standards operating procedures (SOPs) used making it an ideal cohort for an explorative study such as ours, however, our findings need independent and optimally multi-center validation. (II) Despite the single-center nature and the strong SOPs used which should allow for less variability compared to multicenter studies, we identified an unidentified driver of variability in the dataset that resulted in inflated p-values during differential expression analysis. We, therefore, cautiously selected conservative cut-off parameters in identifying differentially expressed proteins that had a p-value $<$0.01 and fold change $>$1.5 and conducted further analyses using a machine-learning selection technique as an additional stringent method of identifying DEPs. (III) Results of this study are purely based on association analyses with an element of targeted bias approach as a limited number of carefully selected target proteins were tested in a single cohort. The biological functions and PD pathology related mechanistic associations inferred from the results generated warrant further future investigations including validation in other larger PD cohorts as well as functional studies assessing the roles and mechanisms of identified proteins.

5. Conclusion

We present a group of 14 stable markers that we identified to be differentially expressed between PD and HC. Of these, four markers: NOTCH1, ALCAM, DUS3, CD36 were also determined to be potential predictive markers for disease progression and the non-motor symptoms in PD such as cognitive functioning (as measured with the MoCA score), and depression (as rated by the BDI and GDS). Both tyrosine kinases CSK and BTK were also found to correlate with baseline levels of total-α-synuclein which in this study were found to be higher in PD cases compared to healthy controls. In addition to these markers, four other markers: Contactin 1, NEGR1, TrkB, and BTK were shown to be capable of predicting a decline in motor abilities as measured by the MDS UPDRS total score, as well as cognitive decline, and depression despite their lack of stability for use as diagnostic markers for PD. Multiple functions of the identified markers have elucidated their involvement in several mechanisms that could be linked to PD symptoms such as cell adhesion, axonal guidance and synaptic plasticity, and neuroinflammation via inflammasome activation and specific T-cell activity. Our findings validate, in part, previous similar studies, and also provide new candidate markers that warrant further validation as well as advanced clinical studies to provide novel insights into the molecular mechanisms involved in PD pathology as a basis of future therapy development.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.nbd.2023.105997.

Ethics approval and consent to participate

Approval was received from the ethical standards committee on human experimentation for all experiments with human participants (approval no. FF89/2008). Written informed consent was obtained from all study participants (consent for research). The study is registered at the German Register for Clinical Trials (DRKS00000540) according to the World Health Organization Trial Registration Data Set.

Consent for publication

Not applicable.

Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Funding

The work conducted by the OE laboratory was supported by the Qatar Biomedical Research Institute under Internal Grant VR98. The funders had no role in the design and conduct of the study, in the collection, management, analysis, and interpretation of the data, in the preparation, review, or approval of the article or in the decision to submit the article for publication.

CRediT authorship contribution statement

Ilham Yahya Abdi: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft. **Michael Bartl:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft. **Mohammed Dakna:** Data curation, Formal analysis, Methodology, Writing – review & editing. **Houari Abdesselem:** Funding acquisition, Project administration, Resources, Writing – review & editing. **Nour Majbour:** Data curation, Project administration, Writing – review & editing. **Claudia Trenkwalder:** Supervision, Writing – review & editing. **Omar El-Agnaf:** Conceptualization, Supervision, Writing – review & editing. **Brit Mollenhauer:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

IA, MD, HA, NM, and OE have no competing interests to report. **MB** has received funding from the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – 413,501,650. **CT** has received honoraria for consultancy from Roche, and honoraria for educational lectures from UCB. CT has received research funding for the PPMI study from Michael J. Fox Foundation and funding from the EU (Horizon 2020) and stipends from the (International Parkinson's and Movement Disorder Society) IPMDS. BM has received honoraria for consultancy from Roche, Biogen, AbbVie, UCB, and Sun Pharma Advanced Research Company. **BM** is a member of the executive steering committee of the Parkinson Progression Marker Initiative and PI of the Systemic Synuclein Sampling Study of the Michael J. Fox Foundation for Parkinson's Research and has received research funding from the Deutsche Forschungsgemeinschaft (DFG), EU (Horizon 2020), Parkinson Fonds Deutschland, Deutsche Parkinson Vereinigung, Parkinson's Foundation and the Michael J. Fox Foundation for Parkinson's Research.

Data availability

Data will be made available on request.

Acknowledgements

The authors thank all patients, healthy subjects, and funding agencies. This work would have not been possible without their valuable contribution. Open Access funding provided by the Qatar National Library.

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